

Portfolio-Conditioned Dynamic Weakness and Reinforcement Ranking in Inverter-Rich Power Networks

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Abstract—When synchronous generators (SGs) are replaced by grid-following converters (GFLs) unit by unit, a planner needs the small-signal effect of the next replacement and of the next reinforcement in the portfolio at hand. We study these marginal effects on the IEEE 39-bus system with a documented GFL model: all 2^9 replacement portfolios at 63 controller policies, 24 of them a preregistered maximin holdout. In 17/19 base-stable holdout policies the same replacement moves the same tracked electromechanical mode in opposite directions in two nested stable portfolios. The effects are small (median 0.013 s^{-1} , about 0.3 % damping ratio) and vanish above 0.05 s^{-1} ; by the classical chain identity each reversal locates an interaction term of the matching sign. A fixed ranking of units makes a materially worse next choice in 0.38 of decisions, but nearly all of these errors are replacements that cross a singularity-type aperiodic instability of the phasor model at high converter share; screened for stability, a fixed ranking errs in 0.065. For branch reinforcement, the re-equilibrated eigenvalue sensitivity of the current portfolio ranks finite reinforcements with median Spearman 0.996 across the new policies, against 0.712 for the frozen-operating-point derivative, 0.818 for a fixed list learned from other conditions and at most 0.445 for static strength indices; it also ranks branch doublings and outages. With a frozen WECC library GFL the ranking method carries over, while the reversals and the branch ranking do not.

Index Terms—Grid-following converters, small-signal stability, eigenvalue sensitivity, grid strength, replacement planning, set functions.

I. INTRODUCTION

GRID-STRENGTH indices summarize how well a network supports converter-interfaced resources. Single-infeed short-circuit ratios yield context-free lists of weak buses; multi-infeed generalizations—the site-dependent SCR [1], the generalized SCR [2]–[4], the generalized operational SCR [5], small-signal strength of all-inverter systems [6] and impedance metrics [7], [8]—are defined for a given set of converter buses and are thus portfolio-conditioned static screens. Modal participation [9], [10] and impedance-based eigenvalue sensitivities [11] rank dynamic criticality.

Retirement poses a sequential question. Units are replaced one at a time, and the planner must decide which unit to replace next *given* those already replaced, and which branch

to reinforce *given* the resulting portfolio. The object is the contextual marginal

$$\Delta_i \alpha_{\perp}(S) = \alpha_{\perp}(S \cup \{i\}) - \alpha_{\perp}(S), \quad (1)$$

the change of the rightmost transverse eigenvalue's real part when unit i is replaced in portfolio S . It has long been known that displacing SGs by converters improves some electromechanical modes and degrades others [12], that rising converter penetration changes the participation in inter-area modes and adds converter-dominated modes [13], that the small-signal effect of converters depends on the mix and dispatch in which they operate [14]–[17], and that a branch reinforcement can reduce linear stability (Braess-type behavior [18], [19]; operating-point-dependent edge weights [20]). Eigenvalue sensitivities with a re-solved operating point are classical [21]–[24], and a recent preprint chains them to line admittances [25]; topology switching changes small-signal margins [26], [27]; and natural dynamic set functions are known to violate sub- or supermodularity [28]–[30].

This paper is an empirical, preregistered study on one benchmark. It contributes four findings, each stated with its limits:

- C1) *Nested same-mode reversal.* The same replacement moves the same tracked electromechanical (EM) mode in opposite directions in two nested stable portfolios in 17/19 base-stable policies of a new holdout. The effects are about one hundredth of s^{-1} and disappear above 0.05 s^{-1} .
- C2) *Why fixed unit rankings fail here.* An oracle fixed ranking errs materially in about a third of next-replacement decisions, but the errors are almost all replacements that cross an aperiodic, singularity-type instability of the phasor model at high converter share; with a stability screen a fixed ranking is nearly adequate, and a portfolio-conditioned gSCR screen is not.
- C3) *Portfolio-conditioned reinforcement ranking.* The re-equilibrated sensitivity of the current portfolio ranks finite reinforcements, doublings and outages better than the frozen derivative, a fixed list learned from other conditions, and static indices; the operating-point term vanishes for controller coordinates and matters for network coordinates.
- C4) *Boundaries.* Single branch actions remove and create EM incompatibilities; under an equal budget one transformer group passes a size-matched null, but only through parameter draws at one policy; with a frozen WECC library

TABLE I
DEVICE AND BENCHMARK DATA (PER UNIT ON DEVICE RATING)

SG	two-axis, $D = 0$, constant P_m , M 4.9–8.4 s (bus 39: 100 s); AVR $K_A \in \{10.1, 40\}$, T_E 0.25–0.95 s by unit; PSS $K_S = -2$, $T_5 = 1$, $T_6 = 4.2$, $T_4 = 0.75$ s
PLL	$k_{p,pll} = 53$, $k_{i,pll} = 1400$
GFL loops	$\tau = 0.03$ s; outer 0.2/8; inner 0.25/6; $x_f = 0.15$, $r_f = 0.01$
Q/V	$k_{p,v}/k_{i,v} = 2/20$, leak $w = 0.05$ rad/s, gain $g \in [0, 1]$
Candidates	$V_9 = \{30, \dots, 38\}$ (2^9 portfolios); core $V_4 = \{30, 33, 35, 37\}$

GFL the ranking method carries over but the reversals and the ranking do not.

The study is phasor-domain small-signal analysis; no electromagnetic-transient claim is made.

II. PROBLEM FORMULATION

A. Model

The network is the IEEE 39-bus system [31] (46 branches, ten SGs, bus 39 an interconnection equivalent) in phasor form with constant-power loads. The system is the index-one DAE $\dot{x} = f(x, z; p)$, $0 = g(x, z; p)$ with device states x , rectangular bus voltages z and parameters p ; at an equilibrium the reduced Jacobian is $A = f_x - f_z g_z^{-1} g_x$. Each SG is a two-axis machine [32] with $D = 0$, constant mechanical power, a first-order AVR $T_E \dot{e}_{fd} = K_A(v_{\text{ref}} + v_s - |V|) - e_{fd}$ and a power-input stabilizer. Machine, AVR and PSS data, ratings and the power flow are taken from the IEEE 39-bus dynamic case distributed with ANDES [33] (GENROU records used in two-axis form, IEEEEX1 gain K_A and time constant T_E , IEEEEST stabilizer); Table I lists the controller values. Each GFL has a synchronous-reference-frame PLL, filtered power measurements, PI outer and inner loops behind a filter reactance, and a leaky Q/V regulator $q_{\text{cmd}} = k_{p,v} g e + x_v$, $\dot{x}_v = k_{i,v} g e - w(x_v - q_{\text{ref}})$, $e = v_{\text{ref}} - |V|$, $w = 0.05$ rad/s. Converter current limits, DC links and delays are not modeled.

A policy $\theta = (g, k, t, h)$ sets the Q/V gain g of every GFL, a common scale k of the surviving SGs' AVR gains, a scale t of their AVR time constants, and a heterogeneity exponent h with $T_{E,i} \mapsto G(T_{E,i}/G)^h$, G the geometric mean. Replacing unit i installs a GFL with the SG's active and reactive dispatch and rating (matched dispatch), so the base power flow is the same for every portfolio. The flagship policy P4 = (0.03625, 1.425, 1.5, 1) makes V_4 unstable with a 0.62 Hz critical mode while all its subsets are stable.

B. Transverse stability

With $D = 0$ and no governors the model has an exact rotation symmetry and a neutral frequency mode (two structural zero eigenvalues). Stability is evaluated on the transverse quotient that removes them; $\alpha_\perp$ is the largest real part of the transverse spectrum. A portfolio is STABLE if all transverse eigenvalues satisfy $\text{Re } \lambda \leq -2 \times 10^{-3}$, UNSTABLE if one has $\text{Re } \lambda \geq 2 \times 10^{-3}$, and unresolved otherwise.

C. Contextual marginals and modal levels

For a policy θ , $f(S) = \alpha_\perp(S; \theta)$ on $S \subseteq V_9$. A marginal (1) is *material* if $|\Delta_i f(S)| \geq \tau_{\text{mat}} = 0.01 \text{ s}^{-1}$ (about 0.2% damping ratio at 0.7 Hz); its sign class is -1 (stabilizing), 0 or $+1$. Because a change of $\alpha_\perp$ can be a jump between modes, each marginal (S, i) is graded: (A) S is STABLE; (B) the critical mode of S has a match in $S \cup \{i\}$ that is the critical mode of $S \cup \{i\}$; (C) B, with both critical frequencies in 0.1–2.0 Hz; (D) for a pair of contexts, C at both, and their critical modes match each other. Matching uses the modal assurance criterion ($\text{MAC} \geq 0.8$) on the 78 rectangular bus-voltage components of the mode shape (unit norm, phase fixed at the largest entry), among eigenvalues in 0.1–2.0 Hz with $\text{Re} \geq -1 \text{ s}^{-1}$ plus the critical one, with $|\Delta f| \leq 0.15$ Hz. Level D is the “same mode” statement.

III. MARGINALS AND SECOND DIFFERENCES

For $S \subseteq V$ and distinct $i, j \notin S$ let $d_{ij}(S) = f(S \cup \{i, j\}) - f(S \cup \{i\}) - f(S \cup \{j\}) + f(S) = \Delta_i f(S \cup \{j\}) - \Delta_i f(S)$, the one-step second difference (interaction term). f is submodular if $\Delta_i f(S) \geq \Delta_i f(T)$ for $S \subseteq T$, $i \notin T$, equivalently $d_{ij}(S) \leq 0$ everywhere, and supermodular with the inequalities reversed [34]–[37]. The following algebra is classical; we use it to locate interactions, not as a new result.

Lemma 1 (chain identity). *Let $S_0 \subseteq V$, $j_1, \dots, j_m \in V \setminus S_0$ distinct, $S_r = S_{r-1} \cup \{j_r\}$, and $i \notin S_m$. Then $\Delta_i f(S_m) = \Delta_i f(S_0) + \sum_{r=1}^m d_{ij_r}(S_{r-1})$.*

Proof: $d_{ij_r}(S_{r-1}) = \Delta_i f(S_r) - \Delta_i f(S_{r-1})$; sum over r . ■

A *nested reversal* of i is a pair $S_1 \subsetneq S_2$, $i \notin S_2$, with $\Delta_i f(S_1)$, $\Delta_i f(S_2)$ in opposite sign classes; its direction is $s \rightarrow d$ (stabilizing at S_1 , destabilizing at S_2) or $d \rightarrow s$.

Proposition 1. *A nested $s \rightarrow d$ reversal is incompatible with submodularity and a nested $d \rightarrow s$ reversal with supermodularity; along the chain from S_1 to S_2 some term satisfies $d_{ij_r} \geq D/m$ if $D = \Delta_i f(S_2) - \Delta_i f(S_1) > 0$ ($\leq D/m$ if $D < 0$), $m = |S_2 \setminus S_1|$.*

Proof: Under submodularity all terms of Lemma 1 are nonpositive, so the marginal cannot increase along a growing chain; supermodularity is symmetric. The m terms sum to D , so the largest is at least D/m . ■

The bound holds for every nested pair by construction; its use is to point at an interaction d_{ij_r} between i and a unit j_r added on the way from S_1 to S_2 . A reversal between *incomparable* contexts does not contradict submodularity: on $V = \{i, a, b\}$ with $f(\emptyset) = f(a) = f(b) = f(ab) = 0$, $f(i) = 2$, $f(ai) = -1$, $f(bi) = 1$, $f(abi) = -2$, all six second differences are nonpositive, yet $\Delta_i f(\{a\}) < 0 < \Delta_i f(\{b\})$. That submodular example also contains the nested $d \rightarrow s$ reversal $\emptyset \subset \{a\}$, as Proposition 1 allows.

Proposition 2 (preference drift). *For a chain as in Lemma 1 and $i, j \notin S_m$, $(\Delta_i f - \Delta_j f)(S_m) - (\Delta_i f - \Delta_j f)(S_0) = \sum_{r=1}^m (d_{ik_r} - d_{jk_r})(S_{r-1})$, with k_r the added units. If the preference between i and j reverses materially, every fixed*

ranking errs in one of the two contexts, and some $d_{ik_r} - d_{jk_r} \geq 2\tau/m$.

Proof: Apply Lemma 1 to i and j and subtract. ■

Remark 1. If $f(S) = h(|S|) + \sum_{k \in S} w_k$ then $d_{ik} = d_{jk}$ and the ranking by w is exact in every context, whether h is concave or convex. Heterogeneity of interactions across units is thus necessary for a fixed ranking to fail; the sign of the interactions is not what matters. Sub- and supermodularity refer to f on all of 2^V ; the stable portfolios do not form a lattice, so no greedy guarantee is at stake.

IV. DYNAMIC INTERVENTION SENSITIVITY

Let $w = (x, z, r)$ collect states, bus voltages and freed references r . Under *scheduled-power-and-voltage* (SPR) semantics the references of every unit are re-solved so that it keeps its scheduled active power and terminal voltage; the slack machine at bus 39 absorbs the loss change and fixes the angle reference, and the frequency is the nominal one, which pins the symmetry directions. The equilibrium solves $R(w; a) = 0$. For a simple critical eigenvalue λ with right and left eigenvectors u, v of A ,

$$\frac{d\lambda}{da} = \frac{v^H (\partial_a A + \partial_w A[\dot{w}]) u}{v^H u}, \quad \dot{w} = -R_w^{-1} R_a \quad (2)$$

(implicit function theorem). The first term alone is the *frozen* derivative D^{fro} ; the full expression is the *total* derivative D^{tot} . This is the feasible-sensitivity construction of [21], [22] applied to portfolio-specific operating points; no novelty is claimed for it.

Remark 2. If a is a controller gain of a unit whose scheduled output is fixed (the GFL gain g_i , a PLL scale, an AVR gain), the SPR states (x^*, z^*) do not move; only the freed references move, and they enter f and g affinely, so $\partial_w A[\dot{w}] = 0$ and $D^{\text{tot}} = D^{\text{fro}}$. Voltage setpoints enter only through references, so their frozen derivative is zero. Branch admittances and loads move (x^*, z^*) .

A branch intervention scales the complete two-port admittance of branch e by γ_e (tap unchanged). The score is $-d\alpha_{\perp}/d\gamma_e$ at $\gamma_e = 1$; the truth is the fully re-solved change $-[\alpha_{\perp}(\gamma_e) - \alpha_{\perp}(1)]$ at $\gamma_e \in \{1.10, 1.25, 1.50\}$. D^{tot} has no fitted parameters, so “held out” qualifies only the comparison baselines. The static baselines are branch flows $|P_e|, |S_e|$; impedance $|z_e|$; endpoint electrical distance on the subtransient bus impedance matrix; effective resistance and Fiedler edge score on the coupling Laplacian L_B ; weighted edge betweenness; endpoint dV/dQ ; and the change of the generalized SCR of the target portfolio at $\gamma_e = 1.5$. The static indices are computed on the base network and power flow and therefore do not change across conditions; we add, post hoc, a data-driven fixed list (the mean finite-effect rank over the conditions of the other clusters).

V. EXPERIMENTAL DESIGN

Policies and holdouts. Policies lie in $u \in [0, 1]$ ($g = u^2$), $k \in [0.5, 2.3]$, $t \in [0.5, 3.0]$, $h \in [0, 2]$. Fifteen *discovery* policies lie on axis-aligned lines through P4; an *old holdout*

of 24 Latin-hypercube points was used in a first campaign; the *new holdout* N01–N24 is a maximin Latin hypercube (best of 2000 designs, seed 20260929) fixed before any computation and never screened for stability. Policies whose all-SG base is unstable (5 new, 6 old) are kept and excluded from tests that need a stable base.

Envelopes. Four envelopes perturb machine groups fleet-wide (EM-f) or per unit (EM-u), converter groups (EC), or both (EMC). Ten fresh Latin-hypercube draws per envelope are evaluated at P4; they are held out in parameters, not in policy.

Computations. All 512 portfolios of V_9 are solved with mode shapes at every policy (32 256 portfolio evaluations). Branch sensitivities are computed at 88 new-holdout conditions for all 46 branches; the implicit-function derivatives match small-step finite differences within 5% (or 10^{-4}) in 100% of 4048 cases. Topology is evaluated at 16 policies for 35 admissible outages and 46 doublings. Reruns of selected tasks were identical in every summary field.

Cross-model holdout. No validated grid-forming model exists in our code base. The one validated alternative dynamic converter is a WECC library GFL chain (PLL, REGC, REEC-B and REPC-A blocks [38] in ANDES [33]) whose parameters were frozen and validated against an EMT implementation in earlier work on a different IEEE 39-bus configuration. We attach it unchanged (constant- Q mode) to a transcription of our machine models that reproduces the internal $\alpha_{\perp}$ of the all-SG base within $2 \times 10^{-7} \text{ s}^{-1}$ at every tested policy (ALT-WECC); the composition itself is validated only at that base and by initialization residuals. Three integrators per converter that belong to disabled or unused control paths and are structurally decoupled in the reduced state matrix are removed before the stability verdict; these rules were fixed after one matrix case had been seen (disclosed in the deviation log).

Statistics and preregistration. Policies and draws are designed points, so fractions are coverages of the tested conditions and intervals are percentile bootstraps over policy (or envelope) clusters. Three secondary tests form a Holm family: T1 (sign test of $p^* < 0.90$, raw $p = 1.9e - 06$, Holm $5.7e - 06$), T2 (paired advantage of D^{tot} over $|P_e|$ minus 0.20, sign-flip, raw and Holm $p = 0.061$ on 6 clusters) and T3 (mixing, below). All gates are effect-size thresholds frozen in the preregistration. Post-hoc analyses requested by internal reviewers are labelled as such.

VI. RESULTS

A. Nested reversals

Fig. 1 shows the strongest level-D nested reversal on the new holdout. At N24, replacing unit 33 in $S_1 = \{31\}$ moves the tracked 0.64 Hz mode right by 0.053 s^{-1} ; in $S_2 = \{31, 34, 38\}$ the same replacement moves the same family, now at 0.74 Hz (MAC 0.97 between contexts), left by 0.042 s^{-1} . Table II gives coverage by level. At the preregistered primary level C, 17/19 base-stable new-holdout policies have a nested reversal (coverage 0.89; the gate ≥ 0.75 passes on the point estimate, and the bootstrap interval 0.74–1.00 reaches below it); the same policies have one at level D. Excluding pairs whose

TABLE II

COVERAGE OF REVERSAL ACROSS THE TESTED BASE-STABLE POLICIES AT $\tau_{\text{mat}} = 0.01 \text{ s}^{-1}$. LEVEL C IS THE PREREGISTERED PRIMARY GATE (≥ 0.75 ON THE NEW HOLDOUT).

reversal type	discovery	old holdout	new holdout
any stable-context pair	15/15	18/18	19/19
A: nested, global $\alpha_{\perp}$	15/15	18/18	19/19
B: nested, tracked mode	14/15	16/18	18/19
C: B, EM-band critical modes	14/15	16/18	17/19
D: C, same mode in both contexts	14/15	16/18	17/19

New holdout by direction ($s \rightarrow d$ / $d \rightarrow s$ / both): level C 9/15/7, level D 6/14/3 of 19. Excluding pairs whose rightmost-eigenvalue gap is below τ_{mat} at any of the four portfolios (post hoc): level C 16/19, level D 16/19. Base-unstable policies (kept, excluded): discovery 0, old 6, new 5.

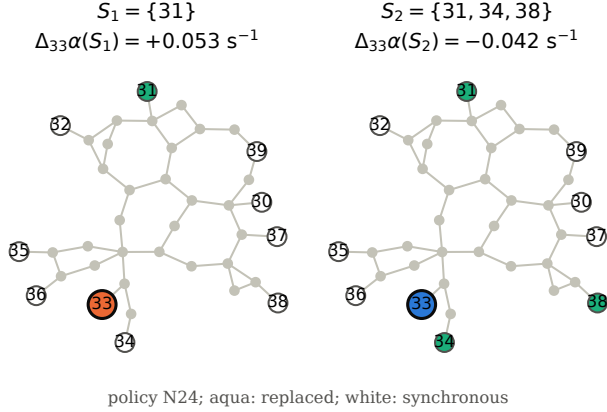


Fig. 1. The same replacement (unit 33) in two nested stable portfolios at new-holdout policy N24 moves the same tracked EM mode in opposite directions (orange: destabilizing; blue: stabilizing). Aqua: replaced units; white: synchronous.

rightmost eigenvalue is within τ_{mat} of the next at any of the four portfolios leaves 16/19. Following the critical mode of S continuously while the added unit is replaced fractionally ends on the MAC-matched mode in 100% of 60 tested endpoint marginals. Fig. 2 shows the three strongest level-D pairs for distinct units.

The effects are small. Over the 3067 level-C pairs of base-stable new policies (87 distinct unit-policy combinations), reversal magnitudes $\min(|\Delta_i f(S_1)|, |\Delta_i f(S_2)|)$ have quartiles $0.011\text{--}0.017 \text{ s}^{-1}$, or $0.27\text{--}0.39\%$ of damping ratio, and the largest is 0.042 s^{-1} . Coverage falls with the threshold (Fig. 3, right): at 0.02 s^{-1} level C holds in 15/19 and level D in 11/19 policies, at 0.03 s^{-1} in 10/19 and 5/19, and at 0.05 s^{-1} in none. Destabilizing $\rightarrow$ stabilizing reversals dominate: at level D, $s \rightarrow d$ occurs in 6 and $d \rightarrow s$ in 14 policies, and both directions on one tracked mode in 3. Converter-parameter draws alone (EC) remove every EM same-mode reversal from the four-unit lattice at P4 (below).

By Proposition 1, $s \rightarrow d$ reversals exclude submodularity and $d \rightarrow s$ reversals supermodularity; both directions occur at level A in 17 and at level C in 7 of 19 policies. The same conclusion follows directly from one-step squares whose four portfolios are stable with EM critical modes: second differences of both signs beyond 10^{-3} s^{-1} occur in 18/19 new and 16/18 old base-stable policies (Fig. 3, left). Fig. 4 maps how often each

TABLE III

FIXED AND SCREENED NEXT-REPLACEMENT RULES (MEDIAN OVER BASE-STABLE POLICIES). ROWS 1–3: AGREEMENT-OPTIMAL FIXED ORDER; PREREGISTERED INSUFFICIENCY BAR $p^* \leq 0.90$ AND REGRET RATE ≥ 0.10 . ROWS 4–6 ARE POST HOC. $\dagger$ FIRST RANKED UNIT WHOSE NEXT PORTFOLIO IS STABLE, ON CONTEXTS WITH AT LEAST ONE STABLE OPTION.

rule	old holdout p^*	old holdout regret	new holdout p^*	new holdout regret
all transitions	0.75	0.37	0.75	0.38
EM $\rightarrow$ EM transitions	0.97	0.03	0.97	0.03
tracked EM mode (level C)	0.98	0.02	0.98	0.02
regret-optimal order, all	–	0.34	–	0.34
stability-screened order $\dagger$	–	0.07	–	0.06
max gSCR of $S \cup \{i\}$ (static)	–	0.77	–	0.76

replacement destabilizes; no unit is uniformly stabilizing or destabilizing.

B. Fixed rankings and their anatomy

The fixed ranking of the nine units that maximizes pairwise agreement with a policy's own resolved preferences is found exactly by dynamic programming. Table III and Fig. 5 report it. Over all transitions it orders 0.75 of resolved pairs on the new holdout and its next choice is at least τ_{mat} worse than the context-aware choice in 0.38 of stable contexts (19/19 policies have $p^* < 0.90$); the regret-optimal fixed order, found by a dynamic program over placed prefixes, reaches 0.34. The old-to-new shift is -0.002 in p^* . A ranking fitted at one policy errs in 0.40 of decisions at another, against 0.38 in sample: the failure is present in sample and transfer adds little (median Kendall between the policies' optimal orders 0.72).

The errors have one source. In 2089 of the 2265 materially wrong decisions (92%) the ranked unit pushes the rightmost eigenvalue out of the EM band; in 22% of the contexts no candidate replacement keeps the portfolio stable. These out-of-band critical eigenvalues are real in 99% of cases, with rates of $20\text{--}3456 \text{ s}^{-1}$ (5th–95th percentile), and appear when five to seven of the nine units are converted. Their participation concentrates on the converters' PLL angle (0.39) and q-axis current (0.36). Replacing the added unit fractionally, $\alpha_{\perp}$ jumps within one 0.025 step of the replaced fraction from about zero to $3773\text{--}109804 \text{ s}^{-1}$, exactly where the smallest singular value of g_z reaches $0.002\text{--}0.007$ (six sweeps, supplement): the signature of a singularity-induced bifurcation of the DAE [39]. These events mark a loss of the index-one property of the phasor model at high converter share; their physical course would require network dynamics or an EMT model, which were not used. Restricted to transitions that stay in the EM band, the oracle ranking orders 0.97 of pairs and errs in 0.03, failing the preregistered insufficiency bar; screening the ranking for stability (first ranked unit whose next portfolio is stable) lowers its regret from 0.27 to 0.065 on contexts with a stable option. A portfolio-conditioned static screen, choosing the unit that maximizes the generalized SCR of $S \cup \{i\}$, errs in 0.76 of decisions.

The preregistered evidence layers (Table IV) meet the frozen rule for the claim that a context-independent node ranking does

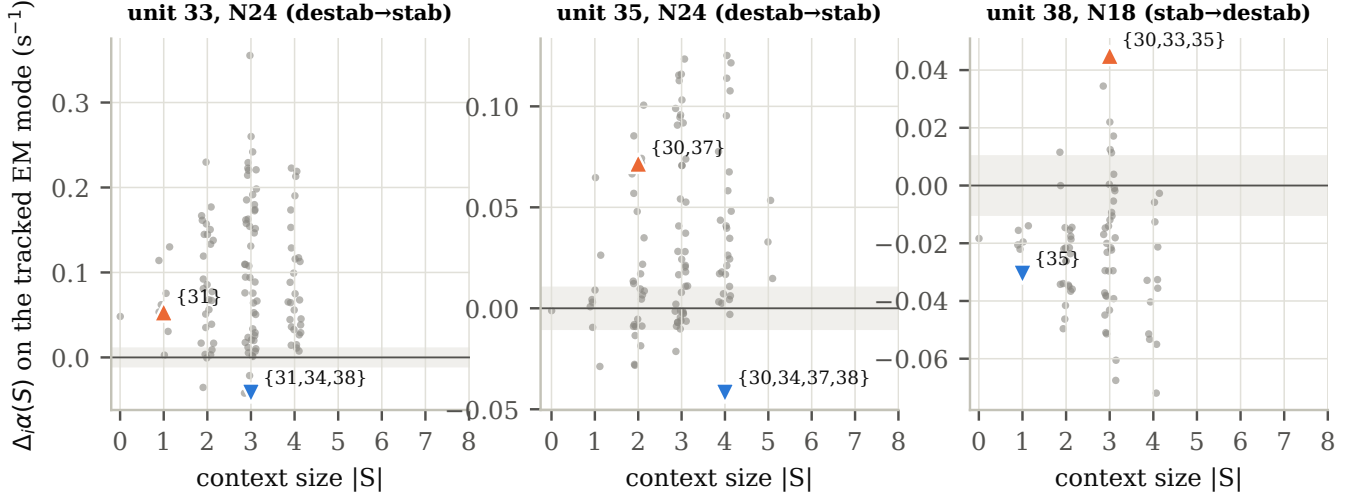


Fig. 2. Strongest level-D nested reversals on base-stable new-holdout policies (distinct units). Gray: the unit’s level-C contexts; triangles: the nested pair; band: $\pm\tau_{\text{mat}}$.

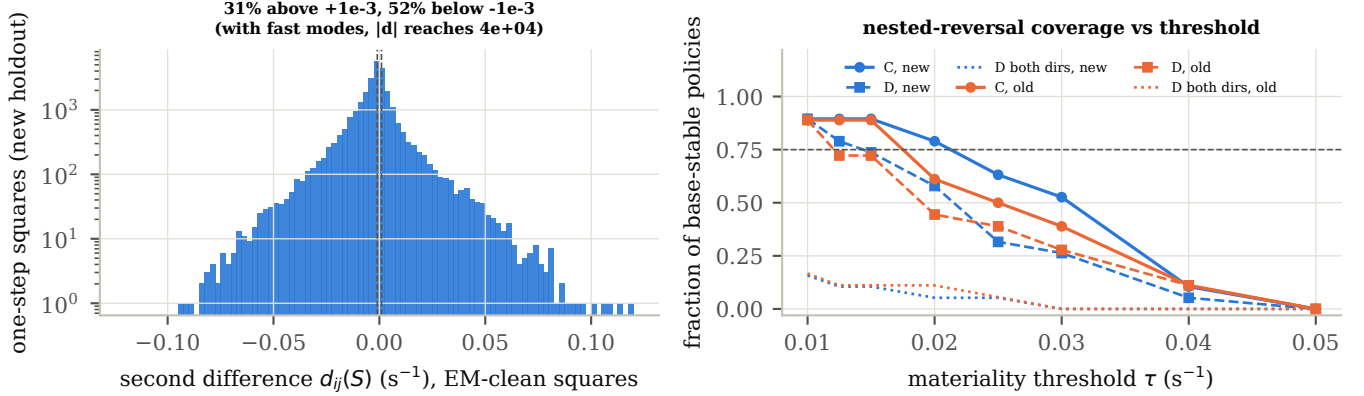


Fig. 3. Left: second differences over one-step squares whose four portfolios are stable with EM critical modes (new holdout). Right: nested-reversal coverage against the materiality threshold (level C, level D, and level D in both directions).

TABLE IV

PREREGISTERED EVIDENCE LAYERS FOR “A CONTEXT-INDEPENDENT NODE RANKING DOES NOT SUFFICE” (NEW HOLDOUT). FROZEN RULE FOR A STRONG CLAIM: ≥ 5 OF 8 INCLUDING L2 AND L5. L1–L4 ARE NESTED, L6 IS POSITIVE WHENEVER L5 IS, L5 FAILS IN THE EM STRATUM, AND L7 USES THE GLOBAL $\alpha_{\perp}$, SO THE LAYERS ARE NOT INDEPENDENT.

layer	value	pos.
L1 arbitrary-context reversal	1.00	yes
L2 nested-context reversal (level A)	1.00	yes
L3 tracked-mode nested reversal (level B)	0.95	yes
L4 EM-band tracked nested reversal (level C)	0.89	yes
L5 optimal fixed ranking failure (FULL)	$p^*=0.749$; regret=0.376	yes
L6 ranking transfer failure	offdiag regret=0.402; diag=0.376	yes
L7 envelope robustness (fresh draws)	EC: 1.0, EM-f: 0.9, EM-u: 0.8, EMC: 1.0; EM mode: EC: 0.0, EM-f: 0.5, EM-u: 0.6, EMC: 0.7	yes
L8 dynamic-converter holdout (ALT-WECC reversal)	0.0 (MODEL-SPECIFIC)	no

not suffice (7 of 8 positive), but the layers are nested and the

ranking layers do not survive the EM restriction; we read the result as: in this model a fixed list fails because it cannot anticipate which replacement crosses the singularity, and within stable EM dynamics it is a reasonable approximation.

C. Portfolio-conditioned reinforcement ranking

Table V and Fig. 6 report branch ranking for the core portfolio V_4 . On the 8 preregistration-eligible conditions (equilibrium residual $\leq 10^{-8}$, a threshold below the solver’s residual of order 10^{-7} in most conditions; six of the eight are draws at P4), the median Spearman correlation with the finite $\times 1.5$ outcome is 0.985 for D^{tot} , 0.913 for D^{fro} and 0.491 for $|P_e|$, with paired advantage 0.492 (interval 0.477–0.608) and top-5 precision 1.000; the preregistered gate passes, while the secondary test T2 does not reach $p < 0.05$ with six clusters. Over all 64 solved conditions (residual $\leq 2.1 \times 10^{-7}$) the medians are 0.987, 0.895 and 0.497.

The informative comparisons are with D^{fro} and with a fixed list. Across the 24 new policies, D^{tot} has median Spearman

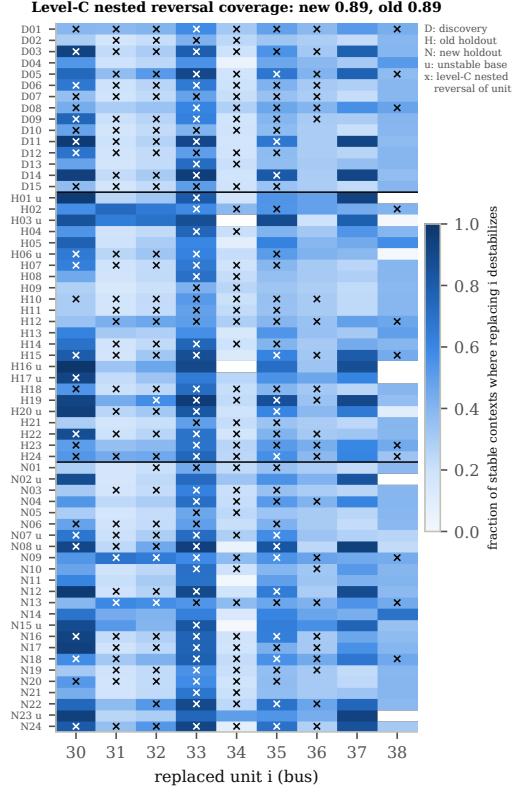


Fig. 4. Fraction of stable contexts in which replacing unit i destabilizes, by policy (D: discovery; H: old holdout; N: new holdout; u: unstable base). Crosses: level-C nested reversal of unit i (white on dark cells for contrast).

TABLE V

BRANCH-REINFORCEMENT RANKING FOR V_4 ON THE 24 NEW-HOLDOUT POLICIES AND THE 40 FRESH DRAWS AT P4 (TRUTH: FINITE RE-EQUILIBRATED $\times 1.5$). MEDIAN OVER CONDITIONS; “ELIG.”: THE 8 PREREGISTRATION-ELIGIBLE CONDITIONS ($R_0 \leq 10^{-8}$); OTHER COLUMNS: ALL 64 SOLVED CONDITIONS. THE PORT FORM RE-DERIVES THE TOTAL DERIVATIVE FROM RE-SOLVED JACOBIANS (ITS “ELIG.” ENTRY USES 2 CONDITIONS). † POST HOC: MEAN FINITE-EFFECT RANK OVER THE CONDITIONS OF THE OTHER CLUSTERS.

predictor	ρ elig.	ρ	τ	concord.	top-5	NDCG@5
$ P_e $ (selected on discovery)	0.49	0.50	0.39	0.71	0.20	0.61
$ S_e $	0.47	0.48	0.37	0.69	0.20	0.61
$ z_e $	0.25	0.24	0.20	0.61	0.20	0.27
electrical distance	0.21	0.19	0.15	0.58	0.40	0.38
effective resistance	0.30	0.30	0.22	0.61	0.60	0.58
Fiedler edge score	-0.05	-0.05	-0.03	0.48	0.20	0.36
edge betweenness	-0.28	-0.27	-0.18	0.40	0.00	0.17
endpoint dV/dQ	-0.17	-0.16	-0.10	0.45	0.00	0.16
$\Delta gSCR(V_4)$	0.16	0.17	0.12	0.57	0.20	0.30
base-portfolio eig. sens.	-0.05	0.03	0.01	0.50	0.60	0.91
frozen (portfolio)	0.91	0.90	0.77	0.90	0.60	0.97
total (portfolio)	0.98	0.99	0.93	0.98	1.00	0.99
port form (verification)	0.99	1.00	0.96	1.00	1.00	1.00
fixed list from other clusters †	0.95	0.94	—	—	0.80	—

0.996, D^{fro} 0.712 and the learned fixed list 0.818; across the 40 draws at P4 the list reaches 0.959 against 0.986. D^{tot} exceeds the list in every condition, but the median paired margin is 0.03 on the draws and 0.17 on the new policies, not the 0.5 gap to static indices, which do not respond to conditions at all. The branch ranking thus changes mostly

across controller policies. D^{tot} also ranks large actions: at 6 new-holdout policies, Spearman with the finite effect of doubling a branch is 0.992 and of removing it 0.885 (top-5 0.80). The first-order prediction overstates the largest reinforcements, whose effect saturates at $\gamma_e = 1.5$ (Fig. 6, right). The port form in Table V re-derives D^{tot} from re-solved Jacobians and is a verification, not an independent method.

D. When re-equilibration matters

Splitting (2) into frozen and re-equilibration parts (Fig. 7): for g_i , PLL scales and AVR gains the two derivatives agree to $2.2e-07$ relative, as the remark predicts. For loads the re-equilibration term exceeds the frozen term in 61% of cases. For branches it exceeds the frozen term for about half of the branches and changes the ranking materially (sign disagreement $\geq 10\%$ or Kendall ≤ 0.6) in 28% of the V_4 conditions; every condition has a branch whose frozen and total signs differ.

E. Robustness to parameter draws

Under the fresh envelope draws at P4 the coverage of a stable-context reversal on the four-unit lattice is 1.00 (EC), 0.90 (EM-f), 0.80 (EM-u) and 1.00 (EMC); for EM same-mode reversals it is 0.00, 0.50, 0.60 and 0.70. The EM reversals are therefore sensitive to the converter parameters. Branch ranking on the draws gives median ρ 0.986 for D^{tot} and 0.503 for $|P_e|$.

F. Cross-model holdout

Table VI summarizes ALT-WECC on the seven preregistered policies with a stable ALT base. Its converted portfolios are less stable than with the custom GFL; V_4 is unstable at all policies. The preregistered test on the global $\alpha_{\perp}$ finds no stable-context reversal, but it is uninformative: in most stable ALT portfolios $\alpha_{\perp}$ is set at -0.100 s^{-1} by the anti-windup tracking state of the plant-level active-power integrator, a state decoupled from the rest of the model that our structural rule kept. On the tracked EM mode, stabilizing replacements occur at 3 of 7 policies and a nested reversal at 1 of 7 (N04), the same policy at which the custom GFL has a level-C reversal on this lattice. Switching on the library’s voltage control (all gains unchanged, post hoc) gives no stabilizing replacement and no reversal, so the data do not support the reading that converter voltage control is what enables reversals. The 12-branch finite ranking is model-specific (median Kendall 0.06), as a previous cross-tool check had indicated, while within ALT-WECC a finite-difference total sensitivity ranks the model’s own outcomes at 0.99 against 0.79 for $|P_e|$, a margin at the preregistered bar. Top-1 agreement of the equal-budget corridors (Section VI-H) is partial (0.71) and the topology directions agree (1.00 on 29 pairs).

G. Topology and electromechanical modes

Topology effects were re-evaluated by tracking the pre-action EM-band rightmost mode of each portfolio through every admissible action (Fig. 8); events in which the full

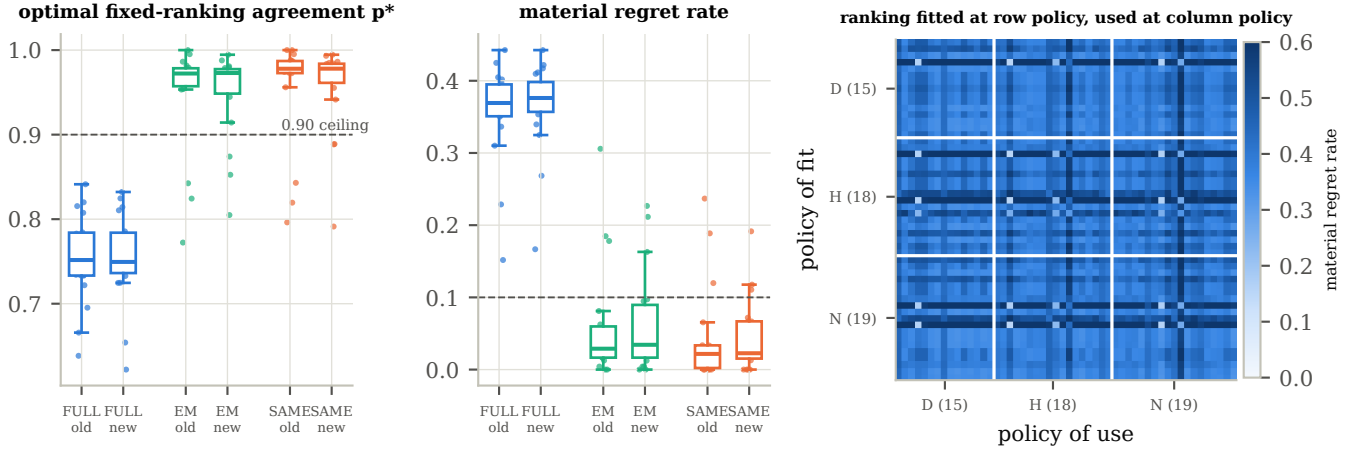


Fig. 5. Agreement-optimal fixed ranking: agreement p^* and material regret rate by stratum (old and new holdout), and regret of rankings transferred between base-stable policies (rows: policy of fit; columns: policy of use, ordered discovery, old, new).

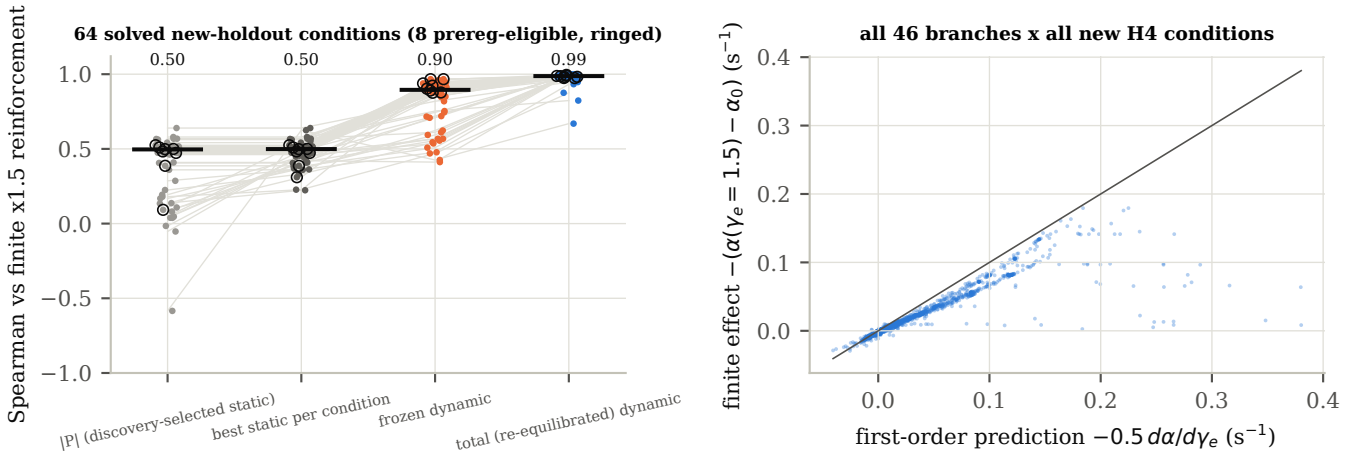


Fig. 6. Left: Spearman correlation with the finite reinforcement outcome per new-holdout condition (lines join the same condition; ringed: the preregistration-eligible subset). Right: first-order prediction vs finite effect, all branches and conditions.

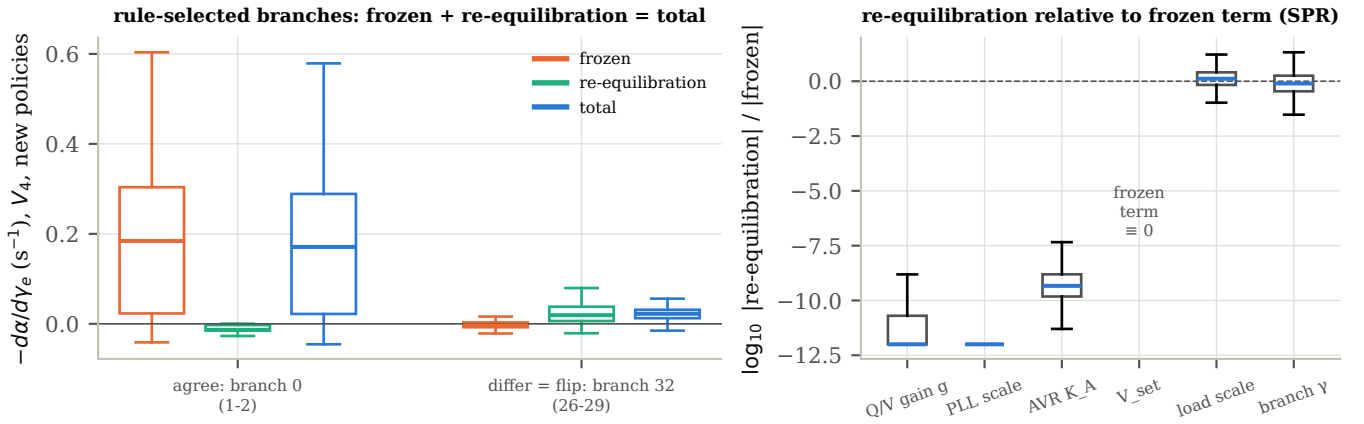


Fig. 7. Left: frozen, re-equilibration and total branch sensitivities for the branches selected by the preregistered rule. Right: ratio of the re-equilibration term to the frozen term by coordinate class (the frozen term of voltage setpoints is identically zero).

$\alpha_{\perp}$ is set by a non-EM mode and differs from the tracked change (50 of 1296 V_4 policy-action cases) are excluded. At P4, doubling any one of lines 1–2, 1–39, 8–9 or the 23–36 transformer makes V_4 stable; across the 16 policies, 45 (policy,

TABLE VI
CROSS-MODEL HOLDOUT: FROZEN WECC LIBRARY GFL CHAIN (ALT-WECC) VS THE CUSTOM GFL ON THE 7 OF 8 PREREGISTERED POLICIES WITH A STABLE ALT BASE. IN ALT THE GLOBAL $\alpha_{\perp}$ OF MOST STABLE PORTFOLIOS IS PINNED AT -0.100 s^{-1} BY A DECOUPLED INTEGRATOR, WHICH MAKES TEST A UNINFORMATIVE.

test	value	verdict
A stable-context reversal, global $\alpha_{\perp}$	0.00 of 7 (custom 0.71)	model-specific
A' same, tracked EM mode (post hoc)	stabilizing at 3/7, nested reversal at 1/7	partial
A'' voltage control on (post hoc)	0/7 global, 0/7 tracked	none
C 12-line finite ranking, Kendall	median 0.06	model-specific
C2 ALT total vs $ P $ on ALT outcomes	0.99 vs 0.79 (3 of 7 below bar)	transfers
D corridor top-1 agreement	0.71	partial
E topology sign agreement	1.00 (29 pairs)	transfers

action) pairs create an EM-failing hyperedge of at most three units, all by outages and all but three at discovery or old-holdout policies (34 and 8). No static topology score meets the prediction rule ($|\rho| \geq 0.6$ in $\geq 75\%$ of policies); their median correlations of about ± 0.5 come from separating outages from doublings, and within doublings they are near zero (Fiedler value 0.00). This is expected N–1 small-signal behavior and we report it as a supporting observation.

H. Equal-budget corridors

A first campaign ranked groups of branches (corridors) by the effect of scaling every member by 1.5, which favors large groups. The preregistered test gives every frozen corridor the same budget $\sum_{e \in C} |\Delta\gamma_e| = 0.5$ and compares its stabilizing effect on V_4 with 500 arbitrary branch groups of the same size (family A; connected groups, family B, as a secondary null). A corridor is called robust if it lies in the top 5% of family A in at least 75% of the 64 new-holdout conditions. Table VII reports the test. The rule is met only by the group of the other eight transformers (the five generator step-up transformers outside V_4 and three network transformers), which is in the top 5% in 0.83 of the 64 conditions (16/24 new policies and 37/40 draws at P4; old holdout 0.78; connected null 0.84). The pass rests on the parameter draws at one policy: across controller policies alone the group would miss the 75% bar. The group of all 12 transformers reaches 0.69, and no spectral cutset exceeds 0.03.

I. Modal mixing is not confirmed

A first campaign reported that controller-weighted modal mixing of the network port operator correlates with the number of reversing units ($\rho = 0.66$ on the old holdout), evaluated at each policy's own critical frequency. At a fixed reference frequency the old-holdout correlation falls to 0.21; on the new holdout the preregistered test gives $\rho = 0.55$ ($p = 0.016$, Holm 0.033), which misses the bar $p \leq 0.01$ (with 19 policies the test has little power at this effect size). In an exact two-factor decomposition of the mixing change, the controller term carries 24% and the frequency term 106% of the variance, with a covariance term of -30%.

TABLE VII
FROZEN CORRIDORS UNDER AN EQUAL BUDGET ($\sum_{e \in C} |\Delta\gamma_e| = 0.5$) ON THE NEW HOLDOUT (24 POLICIES, 40 DRAWS): FRACTION OF CONDITIONS STABILIZED BY $\geq \tau_{\text{mat}}$, FRACTION IN THE TOP 3, MEDIAN EFFECT, AND FRACTION OF CONDITIONS IN THE TOP 5% OF THE SIZE-MATCHED ARBITRARY (A) AND CONNECTED (B) NULLS. GATE: A ≥ 0.75 .

corridor	$ C $	stab.	top-3	effect	null A	null B
other 8 transformers	8	0.67	0.99	0.035	0.83	0.84
all 12 transformers	12	0.65	0.85	0.028	0.69	0.77
step-up transformers of V_4	4	0.51	0.02	0.012	0.00	0.00
cutset 1–2, 3–4, 14–15	3	0.57	0.22	0.023	0.00	0.00
cutset 17–18, 17–27	2	0.00	0.01	-0.000	0.00	0.00
cutset 3–4, 9–39	2	0.53	0.10	0.018	0.00	0.00
line 14–15	1	0.00	0.01	-0.002	0.00	0.00
line 16–19	1	0.66	0.80	0.028	0.03	0.03

VII. DISCUSSION

What a planner needs. On this benchmark, choosing the next unit in context is cheap: nine eigenanalyses per step. The practical content of C2 is therefore diagnostic: a fixed weak-bus list is unreliable because it cannot anticipate which replacement crosses the high-share singularity, and a stability check per candidate—which is exactly an in-context evaluation—removes most of that failure, while a portfolio-conditioned gSCR screen does not. For reinforcement, the result is that the re-equilibrated sensitivity of the planned portfolio ranks reinforcements, doublings and outages reliably, that the frozen derivative degrades across controller policies, and that a list learned from other conditions is good but not adequate across policies. We do not claim a computational saving: our implementation evaluates the total derivative with finite-difference Jacobians at 1.46 s per branch, against 0.61 s for a finite re-solve and eigenanalysis; a saving would require analytic second derivatives and large systems, which we did not test.

Relation to strength metrics and placement. Static indices are computed from the base network and do not respond to the policy or the draw; they carry some information (effective resistance places three of the five strongest branches in its top five) but do not order reinforcements within a portfolio. Placement methods built on grid-strength or Kron-reduced-Laplacian surrogates treat converters as homogeneous and, for grid-forming conversions, can show that each conversion helps [40]–[42]; the reversals here arise with heterogeneous machines and a converter with its own Q/V control, and they were not reproduced by the library converter. Offline priority lists of switching actions [43] are fixed rankings of the kind examined in Section VI-B.

Literature scope. The gap statements rest on a focused search of 2020–2026 journal and conference records, extended to the earlier displacement literature, with arXiv only where no peer-reviewed version exists and without full-text database access; they state the absence of such results within that search.

Limitations. All results are phasor-domain small-signal results on one network with one custom GFL and one parameter box; the holdout tests robustness to controller coordinates, not generality. The machines have $D = 0$ and no governors, loads are constant power, and dispatch is matched; realistic

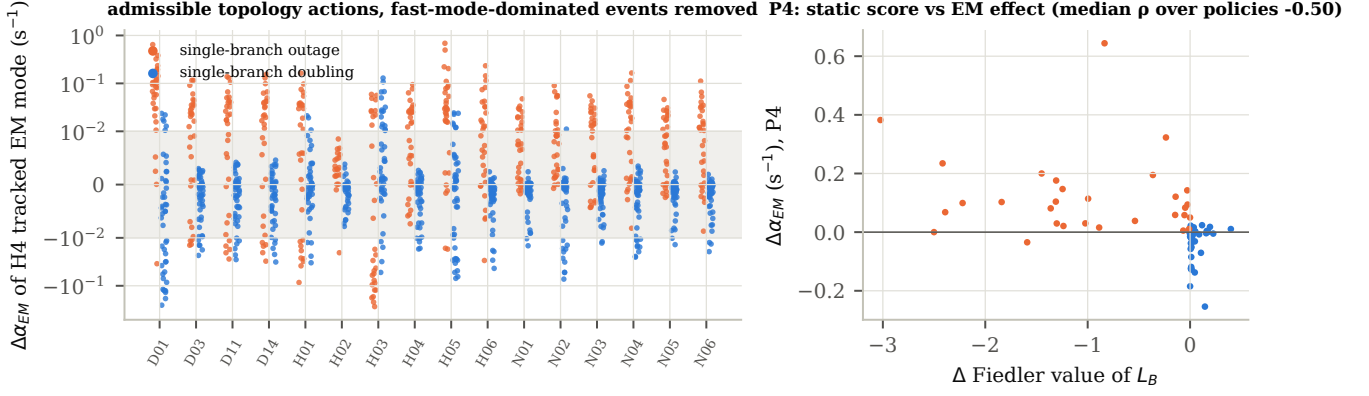


Fig. 8. Left: tracked EM-mode change of V_4 for every admissible action, by policy (fast-mode-dominated events removed; band: $\pm\tau_{\text{mat}}$). Right: at P4, EM effect against the change of the Fiedler value.

damping, governors, voltage-dependent loads or unmatched reactive dispatch could remove the small stabilizing side of many reversals, and they were not tested. The fast events behind the fixed-ranking failure are singularity crossings of an idealized phasor converter without current limits. The EM reversals are sensitive to converter parameters and are not reproduced by the library converter. The primary GOLD-B set has eight conditions because the preregistered residual threshold was stricter than the solver tolerance. The deviation log records execution fixes, post-hoc analyses, a defect in the cross-model tests found by internal review, and hand-written clock times that were later corrected against the commit history.

VIII. CONCLUSION

On the IEEE 39-bus benchmark, the small-signal effect of replacing an SG by a GFL depends on the portfolio in which it occurs: nested reversals on the same electromechanical mode recur on a preregistered holdout, at magnitudes of about one hundredth of s^{-1} . A fixed ranking of units fails mainly because it cannot anticipate aperiodic singularity crossings at high converter share; with a stability screen it is nearly adequate. The re-equilibrated eigenvalue sensitivity of the planned portfolio ranks branch reinforcements, doublings and outages better than frozen derivatives, fixed lists and static indices, and this method, unlike the specific ranking, carries over to a library converter model. Reinforcement rankings should be recomputed for the portfolio and controller settings in which they are applied.

DATA AND CODE AVAILABILITY

Preregistration, deviation log, code, result tables and figure sources are in the project repository, branch `research/contextual-dynamic-weakness-hardening`; software: Python 3.13, NumPy 2.5, SciPy 1.18, ANDES 2.0.

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